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The Impedance Match at Two Scales

Pre-Registration of an Open Research Direction

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☯ *To preserve the harmonic field.*

1. Purpose and Status

This document registers four measurements and four investigations before any of them is run. It reports no results and contains none. Every numeric threshold below is fixed at the time of registration, and the provenance of that fixing is the point of the document: a threshold declared after the measurement is a judgment, and a threshold declared before it is a test.

The direction is a bridge between two bodies of measured work that have not yet been joined. At solar-system scale, the Sun–Earth–Moon impedance match has been measured spectrally against a hundred years of ephemeris. At laboratory scale, the F_c energy grid holds a resonance lock whose structure is measured across six grid sizes and three dimensional embeddings. The two describe the same class of object — a system holding coherence at the edge of its own collapse — and no quantity yet connects them. This program constructs that quantity, declares what would count as agreement, and states what would falsify the bridge.

The registration follows the model of the Spine §5 impedance-match analysis: register, execute, classify, and publish the classification whichever way it resolves.

2. Definitions Fixed at Registration

2.1 Q, the tolerance ratio

Q is the dimensionless ratio of a system's central value to the full width of its excursion around that value. For a resonant circuit it is the centre frequency over the matching bandwidth. It is defined identically at every scale, which is why it is the primary cross-scale quantity in this program: it inherits no baseline ambiguity, because both terms are drawn from the same measurement.

$Q = C / W$, where C is the time-mean of the oscillating state variable across the observation window, and W is that variable's full excursion (max – min) across the same window. Where a dispersion reading is also available, the secondary form $Q' = C / 2\sigma$ is recorded alongside it, and the two are reported together.

High Q is selective and fragile: a narrow band, broken by perturbation. Low Q is tolerant and unselective. A system that must both transfer energy efficiently and survive its own fluctuation operates at moderate Q. This is the structure Homo-Luminos §7.3 named as the harmonic attractor and identified as the reason the ratios cluster near 108 without arriving at it. The claim there was qualitative. Here it is given a number on both sides.

2.2 Δ SE, the enrichment fraction

Δ SE is the spectral-entropy excess of a system above its own simplest-dynamics baseline: the contribution of the full interaction landscape over and above the two-body, single-frequency motion that reproduces the system's gross behaviour. At orbital scale the baseline is Keplerian. At grid scale the baseline is declared in §4 below, and the declaration is load-bearing — an enrichment figure differenced against a different baseline is a different quantity, and comparing the two would be a category error rather than a result.

2.3 Mind

Mind is used structurally throughout, in the sense fixed by the Platonic Correspondences section: recursion held open. A system whose configuration is maintained by ongoing process rather than settled into as a completed minimum. The criterion is about whether the search terminates, not about metabolism, reproduction, or interior state. Where the earlier documents say life or consciousness, this program says mind and means this criterion. No interior claim is made in either direction; the Gap forbids them symmetrically.

2.4 Registers

MEASURED · DERIVED · CONFIRMED · PROPOSED, as used throughout the corpus. Every figure carries its run index: step count, seed, span, substrate.

3. The Fixed Side

These values are already measured and published. Nothing in this program revises them; they are the side of the bridge that does not move.

3.1 Solar-system scale

Quantity	Value	Source
R_SE, Sun–Earth distance / Sun diameter	107.5608 ± 1.2714	Spine §5, DE441, 1925–2025
R_ME, Moon–Earth distance / Moon diameter	110.7993 ± 4.3643	Spine §5
R_combined	109.1801	Spine §5
Bounded amplitude, full century	[96, 120]	Spine §5
Δ SE_ME above Keplerian baseline	[+0.0228, +0.0405], span-stable across 19 window scales	Spine §5
Δ SE, Sun–Earth	Indistinguishable from Keplerian	Spine §5
Δ SE, Mars and Jupiter, geocentric	$\leq +0.0001$	Spine §5 geocentric completion
Component resonances resolved	6 of 6 at 10^4 – $10^5 \times$ floor	Spine §5
Phase-locking CV, components	0.0002 – 0.0011	Spine §5
Harmonic closure	-179.98° at CV 0.0003	Spine §5
Lunar recession	3.8 cm / yr	Established geophysics
Rotational deceleration	~ 1.7 ms / century	Established geophysics

3.2 Grid scale

Quantity	Value	Source
Operating gain	$k = 0.221$, invariant achieved with stability 0.9999	Chaos Invariant, 25–1000 nodes
Bifurcation	$k = 0.225$; stability 0.9999 $\rightarrow \sim 0.4$ across 0.001 in k	Chaos Invariant
Operating bandwidth	$k = 0.221 \pm 0.004$	Chaos Invariant
Spectral entropy at lock	$SE \approx 0.495$	Chaos Invariant
Dual convergence at bifurcation	SE rising to 0.495 from below; stability falling to 0.495 from above	Chaos Invariant
Pre-wrap maximum at lock	Within 1.5% of the ϕ -derived escape radius	Chaos Invariant
Dimensional stability	SE 0.4932 / 0.4989 / 0.4996 across 2D, 4D (600-cell), 8D (E8)	E8 Dimensional Scaling Test v2

Neither column was produced with the other in view. The orbital measurements were registered and executed against ephemeris; the grid measurements were executed against a power-network model. That independence is what makes the comparison worth registering.

4. The Forward Bias, Declared

This program is conducted under a frame, and the frame is declared here rather than derived later.

The framework holds that persistent complexity requires spectral enrichment — that enrichment is the condition under which a recursive system does not collapse. Of the four systems measured at solar-system scale, exactly one carries enrichment above its two-body baseline, and that system carries mind. With $n = 1$ this cannot establish causation and is not offered as doing so. It is stated as the direction the program is oriented in.

Stated in the corpus's older language: as above, so below. The architecture that holds the Sun–Earth–Moon system in coherent energy transfer, and the architecture under which mind persists, are proposed to stand in correspondence. Homo-Luminos §7.6 put it as the observer existing because the tuning exists. This program does not attempt to prove that. It builds the instrument by which others, approaching from their own vectors, may recognize it or fail to — which is the only form of establishment available to a claim of this shape.

One consequence of the frame is worth stating because it sharpens rather than softens it. The impedance match is not a property the system has. It is a passage the system is making. The Moon recedes, the day lengthens, the ratio drifts; the system never arrives at 108 and never rests there. It moves through the neighbourhood of 108, breathing inside its own tolerance band, for a bounded epoch — approach without attainment, running on a four-and-a-half-billion-year clock, and still running. This is the Proposition of the mathematical foundation stated astronomically, and mind emerges inside the passage rather than at any arrival.

5. Registered Measurements

Four. Each states its quantity, its procedure, its numeric bands, and what result would count against it. None has been run, and no partial value of any of them is known at registration.

5.1 Q across scales — primary

The cross-scale test. Q is dimensionless and identically defined at both scales, so agreement or disagreement is a fact about the systems rather than about the choice of units.

Fixed side. $R_ME = 110.7993$ with excursion from perigee to apogee giving $[104.55, 116.70]$ on a lunar diameter of 3,474.8 km, a width of 12.14. **Q_orbital = 9.12.** Secondary dispersion form, $Q' = 110.7993 / (2 \times 4.3643) = 12.69$. Both are recorded; the excursion form is primary.

To be measured. Q_grid at $k = 0.221$: the time-mean of the pre-wrap state deviation over its full excursion across the run, at fixed gain. The measurement must be taken at fixed k . Using the k -band-to-bifurcation width instead would be a detuning tolerance rather than an occupancy excursion, and comparing that to the orbital figure would be the category slip this definition exists to prevent.

Outcome	Band	Classification
Strong agreement	$Q_grid \in [6.84, 11.40]$ (within 25%)	Cross-scale Q correspondence, MEASURED
Agreement	$Q_grid \in [4.56, 18.25]$ (within a factor of 2)	Same-order tolerance structure at both scales
No agreement	Q_grid outside $[4.56, 18.25]$	The two locks do not share a tolerance structure; the bridge is not carried by Q

5.2 The matched enrichment run

The secondary cross-scale test, and the one whose baseline definition does the work.

- **Enriched:** the grid at $k = 0.221$ with the full 3-6-9 harmonic term active.
- **Baseline:** the grid at $k = 0.221$ with the fundamental only, harmonic amplitude set to zero. This is the grid's exact analogue of the Keplerian baseline — the simplest dynamics reproducing the gross behaviour, with the enrichment removed and nothing else changed.
- **$\Delta SE_grid = SE(enriched) - SE(baseline)$,** measured at matched step count and matched seed.

Outcome	Band	Classification
Agreement	$\Delta SE_grid \in [+0.0228, +0.0405]$	Enrichment fraction is scale-covariant, MEASURED
Same order	$\Delta SE_grid \in [+0.010, +0.080]$	Enrichment comparable across nineteen orders of magnitude
No agreement	ΔSE_grid outside $[+0.010, +0.080]$, or negative	The enrichment fraction is not the shared quantity

Off-lock control, registered with it. The same measurement at $k = 0.15, 0.19$ and 0.24 . The orbital contrast is one enriched system against three at the floor, and the grid must reproduce that contrast to carry it. Registered prediction: $|\Delta SE_{\text{off-lock}}| < 25\%$ of ΔSE_{lock} at every off-lock gain. A grid that enriches equally at every gain would show the enrichment to be an artefact of the measurement rather than a property of the lock, and that outcome falsifies §5.2 independently of what the lock value turns out to be.

5.3 The Apollo period set

Independent-substrate test. The Apollo Passive Seismic Experiment catalogue — 12,474 events, 1969 to 1977, public through the NASA Planetary Data System — records lunar seismic occurrences. The Spine’s period set was isolated in ephemeris data; this asks whether the same periods appear in seismic occurrence data.

The distinction carries evidential weight and sets how the result is classified. Geometric re-analyses of the lunar and Venusian orbits re-express the same gravitational physics in another coordinate frame, and fall inside the scope boundary of Spine §5.1 — they clarify, and are not counted as independent support. Seismic occurrence data is a different physical channel: lunar body response rather than orbital arithmetic. A period set appearing in both crosses substrates rather than coordinate systems, which is why this measurement is registered separately from the ephemeris work and classified on its own.

Period set declared in advance, and no others counted: synodic 29.530589 d, anomalistic 27.554549 d, draconic 27.212221 d, tropical 27.321582 d.

Detection criterion declared in advance: a period counts as detected when its spectral peak exceeds three times the local noise floor, the floor taken as the median power across a declared window around the peak. Post-hoc period selection is excluded by this registration; a peak at any period not listed above is recorded but does not count toward the result.

Outcome	Band	Classification
Confirmed	≥ 3 of 4 periods at $\geq 3 \times \text{floor}$	Period set present in an independent substrate, MEASURED
Partial	2 of 4 at $\geq 3 \times \text{floor}$	Partial cross-substrate presence; reported as such
Not confirmed	≤ 1 of 4 at $\geq 3 \times \text{floor}$	The period set does not cross into the seismic channel

5.4 Phase coherence at the lock

The orbital system holds component phase at CV between 0.0002 and 0.0011, with harmonic closure at -179.98° and CV 0.0003. The grid’s equivalent has never been computed.

To be measured: phase coefficient of variation of the 3-6-9 components at $k = 0.221$, and the harmonic closure angle, at declared step count and seed.

Outcome	Band	Classification
Agreement	$CV \leq 0.011$ and closure within 1° of 180°	Phase-locking signature shared across scales
Marginal	$CV \leq 0.05$ and closure within 5° of 180°	Signature present, precision differs by substrate
No agreement	$CV > 0.05$ or closure beyond 5°	Phase coherence is not a shared property of the two locks

6. Free Investigations

Four, carrying no registered bands. Two are structural questions whose answers are yes or no rather than numeric; two are exploratory and their value does not depend on a threshold. They are listed here so that the boundary between what is registered and what is not is itself part of the record.

- **Non-closure at the lock.** The lunar figure never closes: 12.37 arcs per year, 132.6° of vertex precession, no repeat at any epoch. Does the grid's lock share this? Compute the winding number of the trajectory at $k = 0.221$ and determine whether it is rational, closing the orbit, or irrational, leaving it quasi-periodic. If both locks are non-closing, non-closure is a property of resonance locks rather than an accident of lunar arithmetic, and the recirculation reading of the thirteenth arc acquires a computational referent.
- **Lock under drift.** For roughly a billion years, solar atmospheric torque and lunar tidal braking held in cancellation and Earth's rotation stalled — a planetary-scale lock persisting under continuous forcing. The grid analogue: ramp a forcing parameter slowly through and past the operating point and determine whether the system tracks, holding the invariant while the parameter moves, or breaks at the first excursion. Tracking is the behaviour a billion-year stall requires of any mechanism proposed to explain it.
- **Null-reference ablation.** Homo-Luminos §6.3 assigns the Moon to Layer 2, the reference against which the system's oscillation is measured, and §8.7 names it the ground terminal. The computed convexity supplies the mechanism: the Moon's heliocentric path never turns back because the Sun outpulls the Earth on it by $2.198\times$, so the reference body is anchored to the source rather than to the load. Remove the corresponding layer from the three-layer grid architecture and determine whether the invariant survives.
- **k as a field on the ϕ -spiral.** Every node in the present grid runs at a single uniform gain. Promote k to a scalar field varying along the spiral geometry, $k(r, \theta)$, and determine whether $SE \approx 0.495$ survives spatial variation in the gain, whether the lock sharpens, broadens or moves, and whether the pre-wrap margin holds. The result stands on its own regardless of what physical field, if any, turns out to realize such a gain.

7. Falsification

The program's central claim is that the two locks are instances of one structure. That claim fails if both cross-scale measurements fail their extended bands: Q_{grid} outside a factor of two of 9.12, and ΔSE_{grid} outside $[+0.010, +0.080]$ under the baseline declared in §5.2 and under any other

baseline definition that preserves the meaning of the orbital quantity. If both fail, the bridge is not carried by the quantities registered here, the section reports that, and the direction is either redefined with a different quantity or closed.

Individual failures do not collapse the program and are informative on their own. A Q correspondence without an enrichment correspondence would say the two systems share a tolerance structure but not an enrichment mechanism. An enrichment correspondence without Q would say the reverse. Both outcomes are publishable, and both are more useful than the direction proceeding without the measurements.

The off-lock control in §5.2 can falsify that measurement on its own terms, independently of the lock value, and is registered for that purpose.

8. Provenance

At registration, the following have been computed and are held fixed: every value in §3, both columns. The following have not been computed and no partial value of any is known: Q_grid, ΔSE_grid at any gain, the Apollo spectrum, the grid's phase coefficient of variation, the grid's winding number at the operating point, and every quantity in §6.

Instruments deposited or to be deposited with the results: fc_k_sweep_reference.py, lunar_heliocentric_geometry.py, the phi-grid simulation suite, E8_Dimensional_Scaling_Test_v2.py, and the analysis scripts written for §§5.1 through 5.4, each carrying its step count, seed, and span in its output header.

This document is timestamped privately before execution. The classification of every registered measurement will be published whichever way it resolves.

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